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Volume 3 Number 1

Spring 1995

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COLD FUSION TIMES

INK WARS? "COLD FUSION" MAGAZINE FOLDS Two more newsletters?

(New Hampshire) - The third and final issue of the glossy "COLD FUSION" magazine was its last. The post mortem has spawned more hot finger-pointing along with perhaps one, hopefully two, newsletters. Speculation of its demise grew on the Internet and in other reports. Amid relentless rumors of dissolution, accompanied by one-sided, biased stories of what REALLY happened, "COLD FUSION" magazine seems to have had more of a penchant for personality problems than performance.

Perhaps some of our readers might not miss the loss because many articles and reports in that glossy magazine had already been published in the COLD FUSION TIMES and often up to a year earlier. A few scientists might even find the end to the low octane and often duplicate reporting a relief. Despite it all, the story of this technology requires further education for scientists and students. The cold fusion community always needs good objective reporting, even if it arrives in the form of an ephemeral glossy, which in this case, had real potential, if it had not sunk in a stream of alleged greed and low-wattage science.

(Continued page 6)

ICCF-5 ON LINE ICCF-4 PROCEEDINGS HIT PRESS

(Palo Alto, California) As the COLD FUSION TIMES previously reported, ICCF-4 information is available now. About four pounds, several inches thick, of cold fusion literature chock-filled with theory, materials data, engineering, calorimetry, nuclear product information, and much more. The Conference was co-sponsored by the United States Office of Naval Research and the Electric Power Research Institute [EPRI] which is a consortium of the electric and oil companies. There are two sets of documents. The EPRI Proceedings consist of the manuscripts presented at the Conference. Some are not quite ready for publication based upon the presence of errors in a few, or just copies of transparencies in another. [Ed. In fact data from Dr. McKubre was not accompanied by sufficient text in the Proceedings leading to considerable requests, and hence the excerpt on him in this issue (page 10).]

(Continued page 4).

HELIUM - A FUSION PRODUCT

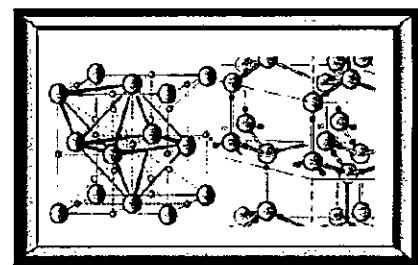
Much more information has been learned about the nuclear products of cold fusion. As discussed in previous issues of COLD FUSION TIMES, helium 4 is one product linked to the production of excess heat. We continue this issue with more data on this subject. (Continued page 7)

CONSERVATIVES HOPE FOR FUSION

(Washington, DC) In September 1994, The House of Representatives had passed HR4908, the Fusion, High Energy, and Nuclear Physics Bill. Within the legislation are provisions which charge the National Academy of Sciences to reinvestigate and then to report on the cold fusion phenomena.

"SECTION 205, National Academy of Sciences Study, states that a report should be prepared to "examine the status and promise of other energy sources, including deuterated metal...."

However, by November the compass of the US shifted right, and on February 7, 1995 the Department of Energy released Budget (Continued on page 9)



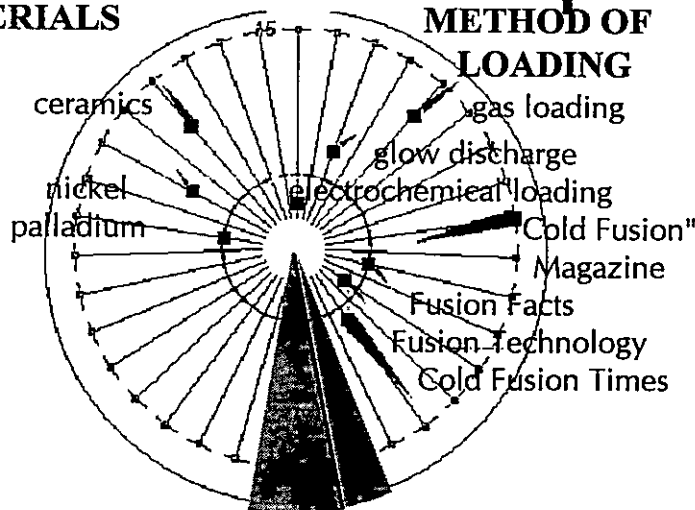
Palladium must be fully filled with deuterons to generate the reactions. The face-centered cubic fully loaded metallic structure is shown on the left, totally filled with deuterons obtained from the heavy water on the right hand side.

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TOWARD THE FUTURE

MATERIALS

METHOD OF LOADING



COLD FUSION RADAR -- Presented by the COLD FUSION TIMES.

The current state of some aspects of the field are figuratively portrayed both by their position (most important - center) and apparent direction.

A BRIEF COLD FUSION UPDATE

by Edmund Storms

The demand for long-term control cells reflects results from the early cold fusion experiments, which produced heat or nuclear products in occasional bursts. More recent work has achieved steady output. Once a cell turns on, it typically stays on for many days. Skeptics who accept the data on excess heat nevertheless assert that it originates not from any nuclear process but from some hitherto unknown chemical reaction. But no evidence from any study has been reported to support various specu-

lated chemical sources. If a novel chemical reaction is the source of this power, significant quantities of some chemical product must be present in the cell, yet no such products have been observed. Moreover, many distinct and novel chemical reactions would be required to explain the excess-heat observations in a variety of chemical environments that have now been studied.

"Experiments that use crackfree palladium and follow the proper procedures now routinely result in excess heat, nuclear products, or both." - Edmund Storms

One reason that cold fusion has been so difficult to accept is that the experiments are so

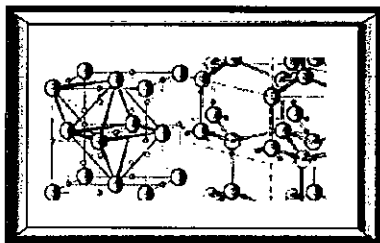
SURFACE AND OTHER EFFECTS

The presence of surface impurities is also important to produce the required deuterium content. McKubre's first experiments, which were successful, used pyrex containers for the cell. When he changed to Teflon, the cell did not work. Further investigation revealed the apparent reason: aluminum and silicon atoms from the pyrex migrated to the palladium surface and

allowed the deuterium to build up to higher concentrations. McKubre's studies also identified another criterion for producing excess heat. His work, which has now been supported by at least four other laboratories, indicates that the electrical current entering the cell must exceed a threshold of at least 150 milliamperes per square centimeter of palladium. For experiments using large pieces of palladium, this can mean a very large overall current.

Many experimenters have failed to produce anomalous effects because they have not applied enough current to their cells. Several experiments have shown that the nature and timing of the electrical input to a cell significantly affects the chances of its success. Heat is more likely to be produced when the electrical current is repeatedly "ramped" from a high to lower level or briefly pulsed to high values.

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[Ed.: We are expanding the TOWARD THE FUTURE section to include the COLD FUSION RADAR. Also here are some comments made by Dr. Edmund Storms about Michael McKubre, who is in the article reviewed on the back of the COLD FUSION TIMES.]

hard to replicate. Many experienced researchers, attempting to follow exactly the methods described in published studies, have, until recently, produced little or no evidence for a nuclear reaction. But scientists are starting to solve the puzzle of why some cells work and some do not.

POWER PROFILE OF THE MONTH

Prof. Keith Johnson

(Cambridge, MA) As Dr. Keith Johnson taught courses on condensed matter and quantum chemistry for the past 25 years at MIT, the 1989 announcement of the cold fusion discovery hardly came as a shock. In fact, Johnson's major area of undergraduate and graduate research at Princeton and Temple Universities dealt with the bonding of hydrogen and palladium and its phenomena. What did come as a shock was the subsequent furor and criticism from his colleagues who insisted cold fusion didn't exist. It was this unrelenting controversy that triggered his creative insight -- not to mention the ensuing reports and studies of the "altered" MIT curves -- that ultimately compelled Johnson to write a moviescript on the subject called "Excess Heat". His work is now in its production phase and has many Hollywood honchos bidding for a piece of the pie.

"I do believe there's a phenomenon. It's of great technological importance."

Johnson was well-prepared to take his pen to paper on the matter. He was not only surrounded by cold fusion politics at a key institution but his vast knowledge of its science made him a natural for letting the world know about both the potential for cold fusion and its "cover-up" by various political and academic officials. Johnson's background led him to examine portions of the physics not generally considered, and he developed a theory at variance with "mainstream" cold fusion thinking. "My immedi-



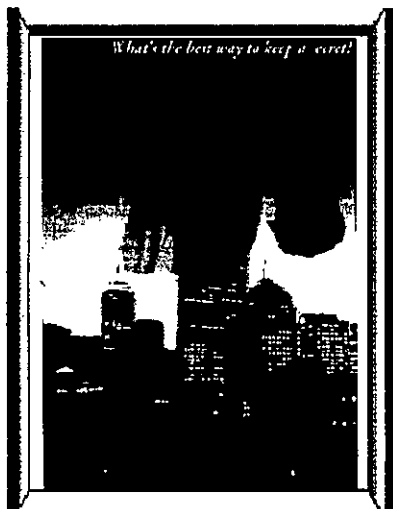
Prof. Keith Johnson

ate response (to the March 1989 announcement) was that the excess heat that they were observing was not nuclear but was connected with internal chemical bonding effects, the same sorts of effects that explained the superconductivity at lower temperatures. The small nuclear effects that were observed were a byproduct," says Johnson. "I believe that the nuclear effects were so small it could not cause excess heat." His theory depends upon the physical structure of the palladium loaded with hydrogen (see cover sheet of this issue). "That's my theorem," says Johnson. "I was always interested in the connection between bonding and solubility. Hydrogen has a tendency to go to the tetrahedral sites. We explained that in 1976 that when you load hydrogen and deuterium in palladium why it becomes superconducting. I do believe there's a phenomenon. And I've always explained that to the hundreds of students I've spoken to about it. It's of great technological importance."

Johnson pointed out that some students are very supportive (of cold fusion), but many of his colleagues are highly skeptical--even though they are confronted with the huge number of documents showing positive results. But Johnson feels that within time even the most hard-nosed critics will one day be believers.

- Gayle Verner

[Ed. - Keith Johnson's theory was presented in the Minsk Conference, the Proceedings of the ICCF-4, and was discussed in the **COLD FUSION TIMES** (vol. 1, number 3). We wish him well in his endeavor in this matter. **EXCESS HEAT** will certainly be a box-office success with perhaps more penetration than the BBC and CBC broadcasts on this subject.]



OF GREAT INTEREST TO OUR READERS SEE FOR THE WORLD (PAGE 8)



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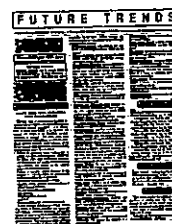
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**Untapped Technology
in Review**



Proceedings: Fourth International Conference on Cold Fusion

Volume 1: Plenary Session Papers

Volume 2: Calorimetry and Materials Papers

Volume 3: Nuclear Measurements Papers

Volume 4: Theory and Special Topics Papers

A second version of the papers was recently published by the American Nuclear Society, a peer-reviewed version. The addresses are on page 6. Because the ANS has been unable to deliver a copy of the ANS version to us for review we rely upon Jed Rothwell's and Dieter Britz's reviews:

"The papers in the ANS version have not been changed or edited much. The English is improved in some, the figures are better, and the printing is better, so the figures and graphs come out a clearer. It is indexed by author and subject. The EPRI version is not indexed at all, which is annoying. The ANS version includes some of the foremost papers from the conference. I was pleased to note that it is dedicated to the memory of Julian Schwinger. Unfortunately, it does not include some of my favorites, like Mizuno, or Cravens. Cravens was voted "best at conference" by Martin Fleischmann. As Tom Passell explained in the Preface, "Not all of the interesting papers were either submitted for peer reviews or passed it, which may say as much about the value of peer review on a radically new topic as about the quality of the research." A few of the ones that did make it into this collection are real duds. Perhaps it should be called "The Best and Worst of ICCF4."

Here is the review with the parallax view of a skeptics through Dieter Britz:

"The editor of FT here explains this special issue, containing 65 (by my count) papers delivered at ICCF-4, Maui, 1993. He mentions a review process, taking more time than expected; thus we can take it that these papers were reviewed. T.O. Passell, of EPRI, who was instrumental in shaping this special issue of FT, here prefaces it with a few remarks. Like the editor, G. Miley, he establishes that the papers were reviewed and that many did not make it through this process, or were not submitted to it. .. Bockris clings to his dendrite theory but not a word of explanation, to be expected after Frank Close's pretty convincing demolition of that idea, being that you can't get keV acceleration energies from something like 20 V cell voltage total. This leaves as the only way out some mechanism for generating a high voltage at the interface, and I don't mean a high voltage gradient. Bush weighs in with not only a transmutation experiment where he finds Cs as a fusion product of Rb with a proton (both isotopes, too, from those of Rb), as well as his theory that wraps up all the mysteries of 'cold fusion' in one fell swoop. He even shows why some people don't find tritium: they used too high a D/Pd loading." [Dieter Britz's files are available on the internet by ftp to vml.nodak.edu, log in as anonymous, giving your email address]

COLD FUSION CROSSWORD #2

Spring 1995

This Month's Puzzle -- A Cold Fusion Anagram.

Solving this puzzle will provide several of the secrets towards successful cold fusion. An example of a solution of such an anagram: One could decode the anagram **MOISTENS LUCID OF** would become **COLD FUSION TIMES**.

- 1) DUPLICATE LOAD HAM ----> _____
- 2) LIMITED OUTER DEUX HI ----> _____
- 3) NEAPOLITAN MUD ----> _____
- 4) TOPIC CREW REEL ----> _____
- 6) GALLON FLUID ----> _____
- 7) EDEMA QUIT TEA ----> _____

ANSWER TO #1:

Winner to #1: Allan Monet, AZ (Best answer closet to fully correct answer) :

A free two (2) Year subscription to the **COLD FUSION TIMES!!!!**

PONS, , SPLIT, WOE
R, , R, , O, , , N, , N
OPTICALPHONON, E
T, , , , A, , , V, OR
INTRAGRANULAR, G
U, , , R, , E, , E, , Y
MAGNETICBOTTLE,
.. A, A, R, C, , , O, A
MASS, , , COMPTON

WHAT'S HAPPENIN' : NEWS FROM THE INTERNET

"There is a particularly fine paper in Volume 2, EPRI TR-104188-V2: M. E. Melich (Navy Postgraduate School), W. N. Hansen (Utah State Univ.) "Back to the Future, The Fleischmann-Pons Effect in 1994." This is a review of some early CF experiments. It is not very technical. Among other things, it describes the 1989 and 1990 Amoco experiments, which were definitive. This is first public mention of these experiments that I have seen, I am delighted that Amoco is finally sharing the data. See pages 10-5, 10-7 and Reference 12.

"In March 1993 the Research Department of the Amoco Production Company provided us with a copy of their internal report of work on the FPE they performed in 1989. . . The result of their 24 Oct - 18 Dec 1989 experiment showed that an FPE experiment in a closed, flow calorimeter produced unaccountably large steady levels of heat, as well as bursts of heat, at magnitudes 100 to 1000 times greater than instrument error. The cumulative net gain in energy was in excess of 50KJ. Furthermore, the tritium level in the electrolyte at experiment start was 2.5 +/- 1.0 pcurie/mL, while at the end of the experiment this had in-

creased to 7.4 +/- pcurie/mL; these results were achieved in a closed calorimeter."

I have seen Ref. 12, and I recommend it: Amoco Production Company, Research Department, T. V. Lautzenhiser, D. W. Phelps, Report T-90-E-02, 90081ART0082, 19 March 1990, "Cold Fusion: Report on a Recent Amoco Experiment."

Another oil company, Shell, has an ongoing cold fusion research effort that is going very well. This is described in the EPRI Proceedings Vol. 1 Paper 9: "Cold Fusion by Sparking in Hydrogen Isotopes: Energy Balances and Search for Fusion By-Products," Dufour, Foos, Millot; and in: J. Dufour (Shell Research S.A., Grand-Couronne, France), "Cold Fusion by Sparking in Hydrogen Isotopes," Fusion Technology, vol. 24, no. 2, 1993, pp 205-228.

- Jed Rothwell

"The Kunimatsu et. al. work is there in several sections. A must for anyone contemplating a "cold fusion" experiment. All those wonderful graphs are there - 19 plots in the first paper alone. Excess heat plotted against everything. But sadly without error bars."

- Tom Droege

[Ed. Although News from the Internet is usually limited to the penultimate page of the COLD FUSION TIMES and COLD FUSION THEORIES are mainly theoretical, in this case because of the importance of this material we are electing to run these interesting excerpts from the Internet.]

PROTON CONDUCTING MATERIALS - PART I

Last issue we reported on proton conductors. That information is supplemented in this issue with more information and additional references. These materials are ceramics and other substances, which often contain rare earth elements and tap into the unusual properties of the hydrogen isotopes within them. We continue this month with more info gleaned from the Internet and other sources.

"A group in Japan at Hokkaido University that claims to get 10+ watts out while putting in less than 5 milliwatts. The whole system is at 300 degrees C so it takes 70 or so watts to get the system up to 300 degrees C and then when they run a small current through the proton conductor, the proton conductor puts out 10 plus watts. Their proton conductor is the size of a quarter and there is a deuterium gas atmosphere surrounding the proton conductor which sits on top of a heater. The proton conductor is so named because it conducts protons (hydrogen or deuterium stripped of the electron) when an electric potential is applied across it.

The group in Japan wrote a paper on their experiment and they described how they made the proton conductor. from their paper: 1) Samples were made from mixing a powder of SrCO_3 , CeO_2 , Y_2O_3 and Nb_2O_5 (Strontium Carbonate, Cerium Oxide, Yttrium Oxide, Niobium Oxide), 2) mixed powder was calcinated at 1440 deg C in air for 24 hours, 3) the powder was then pulverized and filtered to 400 mesh, 4) sample was formed as 1mm thick and 2 cm diameter of coin shape under 10 ton/cm² press machine, 5) sintered at 1460 deg C in air for 24 hours.

Each side the disc (not the edges) is painted with "platinum black" which act as the electrodes and the ceramic disc in the middle acts as the electrolyte - the hydrogen migrates from one side of the disc to the other when the voltage is put across the two faces of the disc. (they now say that they put palladium on the face of the disc instead of platinum) ... The group in Japan does not get the heating effect every time and when they do, it is only with deuterium."

Jeff Driscoll

"A useful rule of thumb is that if the solid is colored or black, it probably won't be a good ionic conductor (electrolyte)."

[Bruce E. Liebert; University of Hawaii]

"(A) proton conductor is an ion conductor with selectivity for protons. Literature sources are abundant since these conductors are the basis for fuel cells, ion selective electrodes (pH being the proton conductor ion selective electrode). Proton conductors come in many forms; polymeric (Nafion), non-hydrogen bonded systems (sintered oxides at high T), hydrogen bonded systems (COO-, glass electrodes).

Selectivities are a function of mobilities, activity coefficients and system equilibrium constants. The conduction can be monitored by a variety of techniques, mostly electrochemical including coulometry, polarization and NMR. You can find more descriptive detail of all of the above in the following books and references within "Superionic Solids and Solid electrolytes, Recent Trends", A.L.Laskar, S. Chandra, eds. Academic Press (1989): includes a chapter on Proton conductors W.E. Morf, "The Principles of Ion Selective Electrodes and of Membrane Transport", Elsevier (1981). and of course any general text on fuel cells. ..."

Cindy Lundgren

"It would seem that the subject of ionic conductors is a fast growing area of materials research with many promising applications. The main and most obvious application of proton conductors is in efficient fuel cells. ...it turns out that proton conductors are not the most efficient nor the most common type of ionic conductor. In fact, one of the first and still one of the best ionic conductors is RbAg_4I_3 with mobile silver ions. The conductivity is approximately 0.27/ohm/cm, which is remarkably high and compares to the conductivity of aqueous H_2SO_4 (35%). Chandra states that:

"In general, both cations and anions can move in a solid lattice. [...] there are expected to be more conducting solids with alkali ions. The possibility of halide ion conductors is small, except for the smallest ion F^- . [...] The more common ions showing high mobility are Li^+ , Na^+ , K^+ , Ag^+ , Cu^+ , F^- and O^- . Most of the compounds of the latter two ions (F^- and O^-) show high conductivity only at high temperatures."

Another class of materials known as "beta-aluminas" has also been extensively studied. These materials have complex formulas like $\text{M}_2\text{O} \cdot x(\text{Al}_2\text{O}_3)$ where $\text{M} = \text{Na}, \text{K}, \text{Rb}, \text{NH}_4, \text{Tl}, \text{Ag}$ and x is approximately 5 to 11. Sodium beta-alumina was the first material found with a conductivity of approximately 1.4×10^{-2} /ohm/cm.

Chandra writes:

"Since then a large number of beta-aluminas with different alkali ion compositions have been synthesized and studied. A large number of other structures have also been obtained but still beta-aluminas are the best alkali metal ion conductors."

Bill Page

1995 MIT IAP COLD FUSION REVIEW Cambridge, MASS. - January 21, 1995

They came from Hawaii, New York, Connecticut, Maine, NH, Utah and even James Griggs from Rome (Georgia). This seemed like an older, more scientifically sophisticated crowd. There was less material science and engineering than there should be (arbitrarily measured by the number of blackboard-surfaces exhausted per microcentury), but a good solid continuation of scientific education at the Massachusetts Institute of Technology. Several hundred people attended the IAP seminar.

(continued on page 8)

The COLD FUSION TIMES apologizes for delay in publication. An intense lightning storm and the difficulty of continuing in this field have slowed us down. A substantial fraction of our equipment has been broken, much now replaced. But to continue we need your subscription to, and support of, the only COLD FUSION journal which is devoted exclusively to the material, economic, and intellectual property issues beginning with the loading of the isotopic fuel - a hydrogen isotope - into a material to produce useful energy.

INK WARS? (from page 1)

"COLD FUSION" magazine was owned by Wayne Green of N.H. and Charles Becker, of Texas. Wayne Green, W2NSD is a prolific publisher who founded "BYTE" magazine, and has sustained '73" magazine for 33 years. Green made Eugene Mallove the Editor. There were problems early on for the glossy mag. By June 1994, internal disagreements between Mallove, some of the other editorial staffers, and Wayne Green reportedly reached crescendo levels.

THE PATENT PAGE

Analysis of the Intellectual Property Field in Cold Fusion

We take a break from the usual patent stuff to the other side of intellectual property.

NEXT ISSUE: Japanese patents.

What happened? In addition to cross accusations between the editor and publisher of the glossy, and the reasonable problems of planning for cash flow needed to actually sustain the glossy, in the most recent issue (Feb 95) of "73", Green purports that more than a "couple of million dollars" is missing. Rumors from various camps have also arisen saying that some of the glossy's staff were only interested in short-term profits including floating up the worth of stock rather than helping the fledgling field. Consistent with this, said Green, is that Becker jettisoned at least one member of the magazine who was connected with another company in the field, just to keep the company's name off the masthead. Amidst such monkeybusiness, the interne-cine fighting naturally grew and with it the end of "COLD FUSION" magazine.

Wayne Green was interviewed by W5YI Magazine (The Nations Oldest Ham Radio Newsletter). Green [W2NSD in

"HE SAID/SHE SAID" COMICS

-- The State of "Cold Fusion" magazine --

W5YI] : " ... we found that the projection that was based on the information from our partner missed by about 90%. There was no choice but to back off and stop publishing ...spending \$30 thousand dollars a month on a glossy magazine."

The other major backer, Charles Becker, a benefactor of ENECO, apparently slipped off the radar screen after faxing a letter from Green to him on to Mallove.

Green: "I have not heard anything further from (Becker) other than he faxed my letter to him to Mallove who then sent me an abusive and I consider libelous letter which he sent to a number of people."

Ignore that "COLD FUSION" magazine was a knock-off copy of the COLD FUSION TIMES. Ignore that most of what was in "COLD FUSION" magazine was in COLD FUSION TIMES months and often years before. Here is the story of what is happening "Down East".

Then, it is reported that Mallove and the others allegedly removed the submitted manuscripts, with both sides then claiming ownership of some part of the paper, and papers, and paper's name, but with little to no mention of the receipts.

Green: The result of that was that I had no choice but to fire (Mallove) at that point. I have asked for him to return approximately 100 pages of manuscripts that he has got which he has not done in a month now."

From Green's point of view, it was Mallove's unrealistic salary demands which, he alleged, contributed to the demise of the

magazine. Over the Fall, Mallove attributed this "divorce" to purported severe managerial and tax problems arising from the offices of Wayne Green, Inc. (WGI), Peterborough, NH. With well-honed PR skills, Mallove urged others to action.

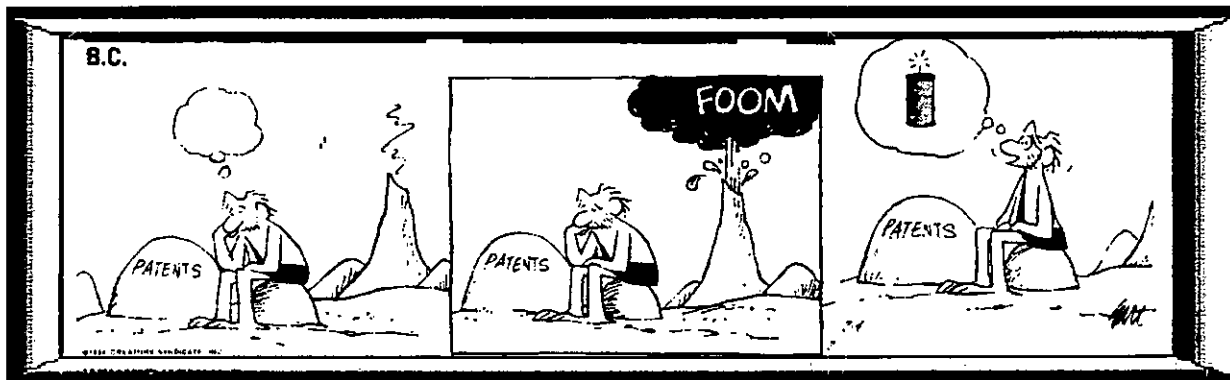


NO MOA

Mallove: "the responsibility for subscription fulfillment resides with publisher Wayne Green, Inc. Subscribers who are dissatisfied with the content or format of the newsletter and its substitution for the magazine should take up the matter with Wayne Green, Inc. or take such other action as they consider to be appropriate."

Meanwhile, Green was determined and has published a substantial newsletter, beginning with a September issue. It has developed into a good newsletter except for substantial weakness in the physics department. In response Mallove ridiculed the newsletter worldwide. Mallove, on Sept. 12, 1994, broadcast on the Internet terse but denouncing comments about his previous publisher.

(continued on page 9)



OUR
FAVORITE
CARTOON
OF THE
"YEAR"

by Auth from
B.C.
Boston Herald
(July 4, 1994)

ENGINEERING AND MATERIALS VIEW TOWARDS COLD FUSION

HELIUM-4 (He^4) PRODUCTION DURING COLD FUSION REACTIONS (Part II) Dr. Mitchell Swartz

The cold fusion reactions are quite novel because they only occur in a physical dense material and not in a "gas-like" plasma. Why is this fusion different? These nuclear phenomena appear to occur as a special case of first-row interstitial lattices fully loaded often to 1:1 atomic ratios within another (e.g., palladium or certain other Group VIII or rare earth) lattice.

Last month we discussed the uniqueness of the palladium hydride phonon system. As a result of this unique quality, there have now been scores of experimental confirmations of new reactions producing both nuclear products and thermal energies (called excess enthalpies) created by

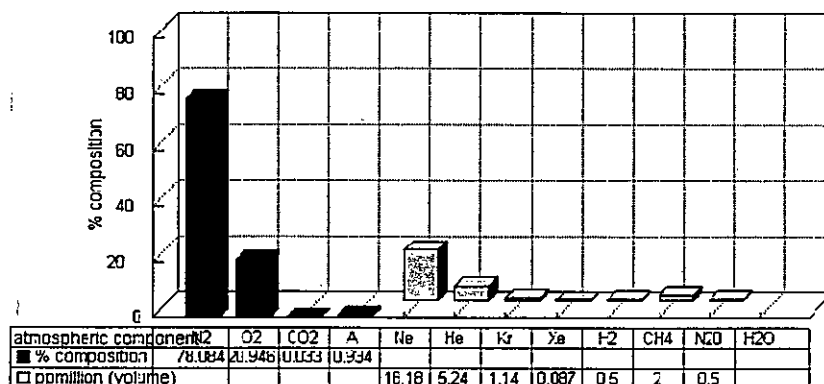
MATERIALS CORNER -- This issue: HELIUM PRODUCTION IN COLD FUSION PHENOMENA

these cold fusion phenomena. These are one of the major products (or nuclear "ashes") of this new technology.

This column (continuing next month on helium interference issues)

will briefly review some of the remarkable findings which clearly demonstrate that the helium-4 "ash" production by cold fusion is consistent with some of the reactions having a probable nuclear origin.

Composition of Atmosphere



This bar graph shows the atmospheric composition for several materials. These are dry data, volumetric. The H_2O , D_2O , DHO , HTO , (etc.) and their complexes, clathrates and stereoconstellations are critical.

HELIUM-4 GENERATION FROM ULTRASOUND

[Readers of the CFT have been aware of the use of ultrasound to generate cold fusion reactions. We augment this issue with comments released by Eugene Mallove on the Internet which describe the E-Quest cold fusion experiments.]

"1. E-Quest now has verified helium production in excess heat-producing experiments that range from 50 watts excess to 500 watts excess. Experiments are done in their heavy water circulation system. Typically Pd is used in these experiments, though Ti works too, but gives a somewhat different response. They do not get excess heat in ordinary water or with stainless steel targets. The excess heat-producing runs to generate helium ranged from 3 hours to 72 hours duration.

2. Helium measurements have now been successfully made with gas samples from the E-Quest experiments --- samples were measured at the US Bureau of Mines (Amarillo, Texas), at SRI International, and most recently at Rockwell (the latter lab is known to be used for helium measurements by investigators in the hot fusion program).

3. Ten (10) runs have been analyzed for helium. ALL gave positive helium results. Earlier runs tended to be in the 50-60 ppm of helium⁴ range. At least one sample measured at Rockwell recently was a remarkable 552 ppm. The error range: plus or minus 1 ppm. E-Quest believes this latter result accounts for nearly all of the excess heat evidenced in that run. This sample, as in other recent testing, was collected in 50cc *stainless steel* bottles. All the sample was obviously not needed for the testing, so archived gas remains for future testing. That 552 ppm sample was collected in a test of the E-Quest device at Los Alamos National Laboratory. At LANL, checks were made for gammas, neutrons, and tritium. None of the latter were found.

4. The cover gas used in all cases was analytical grade argon *not* D_2 . Helium⁴ content of the argon was measured and found to be less than 0.4 ppm. Remember, atmospheric He^4 concentration is less than 6ppm.

5. Among the most amazing claimed findings however, was the highly significant

enhancement of the measured helium³/helium⁴ ratio, which is normally on the order of 10^{-6} . The helium³ measurement at Rockwell showed this ratio to be enhanced to around 10^{-3} -- about 1000-fold increase. Such a finding would virtually rule out any possibility of the helium⁴ being from "contamination" -- as preposterous as that favorite skeptic's theory would seem anyway at this point. Sixty ppm is 10 times natural background helium⁴, but 552 ppm is on the order of 100 times background.

6. Evidence was obtained for plenty of helium in the Pd metal too.

7. The mass-spec peaks defining helium⁴ and D_2 -- or possible "di-deuterino" gas-- are said to be very well separated. So this IS helium⁴.

8. Multiple measurements were made at Rockwell on each of several gas samples and background gas samples submitted."

Also this issue:
PROTON SEMICONDUCTORS- page 5
MORE MATERIAL INFO -- pages 2,4,5

Concerning McKubre --- Bishop (Continued from page 10)

"They have found that if the palladium can be loaded with at least 95 atoms of deuterium for every 100 atoms of palladium, it will produce anywhere from 3% to 50% excess heat. The Electric Power Research Institute finds the heat results firm enough to keep up its support of the SRI experiments in case they are tapping some new energy source. "The heat intensity is at least 10 times too large to be resulting from chemical reactions or chemically stored in the cell," says Thomas O. Passell of the institute's nuclear-power division, who is monitoring the SRI experiments. The immediate problem to solve, Dr. McKubre says, is figuring out why some pieces of palladium will readily soak up the necessary number of deuterium atoms and other pieces of the metal won't. Once the researchers can reproduce and control the heat-producing experiments, they will then try to determine whether the heat is coming from nuclear reactions.

"The SRI scientists also are among those who have seen "heat after death," a phenomenon first described by Drs. Pons and Fleischmann. When the researchers turn off the electricity of an experiment, the hot palladium rod usually cools down in a few hours. In some instances, however, the palladium has stayed hot for as long as three or four days without any electric power pouring into it, as though some internal heater were still at work."

1995 MIT IAP -- January 21, 1995 (Continued from Page 7)

The attendees heard James Griggs share some of his data, as recent as this week, holding part of his bivalved rotor in hand. Keith Johnson shared movie updates alternating with his Jahn-Teller animation showing, and some discussion of, deuteron and hydrogen anharmonic activity and recombination on nickel surfaces and in palladium. Peter Hagelstein discussed his latest updates on neutron transfer and surveyed the recent understandings of screening the coulomb potential. Ray Conley brought his wet cell, heralding the lack of a laboratory-room camera and a towel obvious this year. There were more videos, data, material science, theories, magazines, papers, conversations, and people sharing thoughts and business on cold fusion than sci.physics.fusion's resident TB-skeptics could probably shake their sticks at, even with a few reinforcements from CERN and DOE. Although there were more purported over-unity devices discussed than this electrophysicist could bear (leading to the premature exit), the content of both the cold fusion portion and the interest that the crowd maintained were quite good, thereby propelling another ten score alumni and scientists one full quantum ahead.

Dr. Vesselin Noninski was scheduled to speak, did not, and offers this review: "I thought it might be interesting for some of you to hear one more impression regarding the MIT "cold fusion" day. Probably, I should start at once with stating that, although it was quite enjoyable a day and Gene did a great job, organizing it, this seminar had almost no scientific merit. In most of its part this gathering resembled a promotion fair. As any such fair, guided by salesmanship rather than objectivity, there was an obvious bias towards what was promoted as positive results and any criticism in such a forum appeared out of place. Unfortunately, I remained with the impression that most of the

(continued next page)

FOR THE WORLD (MORE INFORMATION -- SEE PAGE 3 TOO)

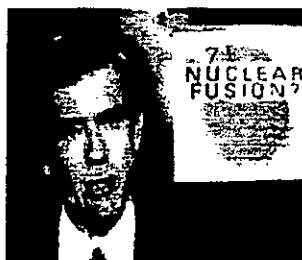
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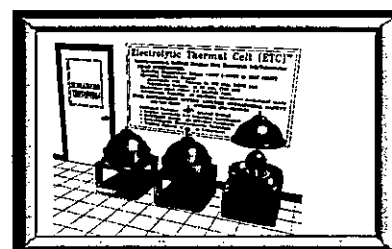
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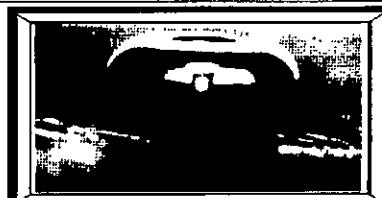
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Wayne Green

(continued from page 6)

Mallove: "(We) have reviewed a copy of this newsletter and consider it to be entirely substandard in its presentation of information on the field of cold fusion. There is no designated editor of this newsletter, so we assume that Wayne Green is himself acting as Editor."

"A more serious matter is the content of Wayne Green's lead editorial, 'So where's your big glossy magazine?' In this editorial, the publisher makes assertions that we believe to be contrary to the truth of the magazine's publication history, and makes untrue and defamatory statements about individuals in the cold fusion field."

And Green's response? From the pages of 73 (February '95). "I won't cry on your shoulder about the missing articles the ex-editor (Mallove) took with him, the stolen mailing lists, and things like that. But it was a mess and the three teams of lawyers involved will end up doing very well by the time all this is over. We're still trying to find where a couple of million dollars went".



1995 MIT IAP -- January 21, 1995

(Continued from Page 4) "participants do not understand that the problems this field is facing are scientific and not technological. I must admit that I was quite impressed by the Griggs' presentation mainly probably because I saw a serious man of action -- a perfect example of an entrepreneur. I cannot afford also to not notice people like Fred Jaeger of ENECO preying on the poor inventors, trying to buy them for peanuts. It is always amusing to hear his frank exposure of the policies of his company. One has to be really quite desperate (like these guys from Russia -- the permanent target of ENECO) or not in his right mind to get into the deal offered by ENECO. One other entrepreneur I probably should mention was a fellow from Hawaii presenting an elaborate scheme of constant magnets coupled with an electric motor or generator. One talk I enjoyed was Peter Hagelstein's. He now is proposing basically two approaches to the problem. One is very similar to the way Raman effect or laser action is explained. He now speaks about a phonon laser effect that can explain the experimental data. The other approach very much resembles Mossbauer effect type of an explanation." (V. Noninski)

(continued from page 1 - Politics) Highlights as part of the Department's FY 1996 budget request to Congress. . The request for Fusion Energy for FY 1996 is \$366,045,000 (1995 \$328,638,000).

"Our research in fusion energy is aimed at developing an inexhaustible energy source for the future. ...Fusion continues to offer the promise of a safe, environmentally attractive, secure, and competitively-priced source of energy. Operation of fusion power plants would not contribute to acid rain or global warming problems and the deuterium fuel required for fusion is plentiful and could be readily obtained from sea water ..."

The FY 1996 budget request focuses on ignition physics, fusion nuclear technology, magnetic confinement configuration optimization, and low activation materials development. The US Congress should pass a fusion bill which authorizes hydrogen, high energy, and nuclear physics research. Is a reinvestigation and report on cold fusion in store for Congress after the 100 days? Conservatives hope so. Liberals and Independents hope so as well. Write your congressperson to amend all legislation and include the provisions for investigations of 'deuterated metals'.

Meanwhile the FY 1996 budget request announced \$29.6-billion over 5 years, down \$2.2 billion by trying to bring spending under control, for stabilizing and cleaning up toxic and highly radioactive waste. Because this remains the largest single environmental problem in the world, and could take decades, and \$250 billion, all solutions to improve this are important (see below).

LETTER CORNER

Rep. Karen Clark was reelected in Minnesota. Here is Dana Rötgerd's letter to Representative Clark shared with us on cold fusion and its importance to resource management and the nuclear waste problem.

Wed, 21 Sep 94

Dear Representative Karen Clark:

I attended the recent DFL convention in St. Paul and staffed a table for the Minnesota Cold Fusion Alliance, a group I and several technological futurists started last February, 1993. We had a small demonstration project going on our table on the ramp. It was a rare earth magnet cold fusion experiment that had produced significant excess heat in experiments run privately from February to April of this year.

The principal thrust of applied cold fusion research is to come up with a commercially viable, simple ~20 Kp fusion reactor for alternate energy application, (probably first in Toyota or Fiat electric cars). Replacing fission power with this technology is this alternate energy goal.

There is solid experimental and theoretical scientific work that one can access through the Fusion Information Center of Salt Lake City, and EPRI's Fourth International Conference on Cold Fusion that suggests that there may be a direct solution to the nuclear waste problem. Simply put, it may be possible a few years from now to dissolve wastes from fission nuclear reactors and using cathodes saturated with hydrogen, add a proton to radioactive isotopes changing, say Cesium 137 to Barium 138.

For the Minnesota legislature to take advantage of this new science is one of my hopes and the reason several prominent Minnesotans formed the Minnesota Cold Fusion Alliance in 1993. Hearings on this area of science could produce some star witnesses both from the local community and some impressive technical demonstrations from the groups who have been working with the University of Minnesota Department of Materials Science. This line of research is controversial (as is all cutting edge science), but possible benefits are staggering. Sincerely, Dana Rötgerd

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COLD FUSION TIMES

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TO COLDLY GO WHERE NO ONE HAS GONE

MORE CF REPORTS:

From time to time a few courageous American reporters examine and report on the cold fusion evidence which they have observed. We salute Alice Waugh at *MIT Tech Talk* [Nov 9 1994 entitled "Graduate Student Envisions Power for Spacecraft from Cold Fusion" (page 8) and Jerry Bishop for his continuing reports on cold fusion in the *Wall Street Journal*. A few brief excerpts demonstrate the importance of the Journal and its open-mindedness about this field.

CONCERNING McKUBRE

Brief excerpts from Jerry Bishop, *Wall Street Journal*, 7/13/1994

"...The lure and the frustration of the cold-fusion experiments are evident at SRI International. Here a team of 10 scientists and technicians under Michael C.H. McKubre is working with a dozen or so cold-fusion "cells," sealed steel cylinders a bit larger than thermos bottles, containing the palladium rods, platinum wires and the heavy-water electrolyte common to many cold-fusion experiments.

"Like everyone else who plunged into cold fusion, Dr. McKubre initially had trouble repeating experiments. An experiment that stayed maddeningly normal for weeks would suddenly spring to life for no observable reason and stay extra hot for hours or days."

"Such unpredictable events plague - and fascinate - researchers. Recently, for example, one SRI cell sat quietly for 2.5 months producing no more than the 50 or 55 watts of power it was consuming. Suddenly, a pump that was keeping the heavy water flowing through the cell at a constant temperature broke. The water near the electrode rapidly heated above the boiling point. This accidental heating of the palladium caused it to begin producing eight to 16 watts of excess heat. Why this happened is a mystery, Dr. McKubre says. But several times since then, the SRI researchers have managed to trip a cell into producing excess power by "cycling" the temperature of the heavy water up and down. The SRI scientists also are learning the conditions necessary for a cell to produce excess power.

(Continued page 8)

"As far as I'm concerned, there is a phenomenon to be studied. There is a mystery. I have observed something that I cannot explain, and I will continue to work until I understand it. ... The issue of reproducibility, or its lack, has been frequently misunderstood in this field. The difficulty has been reproducing the conditions which produce the results, not reproducing the results. So if we meet the conditions that we have established then we, in every case, observe excess heat."

Michael McKubre [BBC Transcript Horizon, 1994, "Too Close to the Sun"]

COLD FUSION TIMES

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